

VendingVision

Confidential · Summer 2026 · For internal use only

THE OPPORTUNITY

PROBLEM	SOLUTION
Vending machine operators running older machines have zero data visibility. They drive to each machine to manually eyeball inventory — no sell-through data, no low stock alerts, no revenue tracking.	A retrofit hardware device that mounts inside any existing vending machine. A wide-angle camera tracks inventory automatically. The owner sees live data on their phone — no new machine required.

BUSINESS MODEL

HARDWARE	SUBSCRIPTION	TARGET CUSTOMER
\$200 One-time device purchase	\$15/mo Per machine, app access	Solo ops 2–10 machines, side hustle

HARDWARE SPECIFICATION

What you're building

A self-contained retrofit device that installs in under 30 minutes, requires no modification to the vending machine, and runs autonomously after setup.

Compute unit	Raspberry Pi 5 (4GB RAM)
Camera	RPi Camera Module 3 Wide (120° native) or fisheye adapter for 180°
Power	Internal tap from machine's 120V AC via 5V USB-C adapter
Connectivity	Built-in WiFi (Pi 5), cellular SIM hat as v2 option
Storage	64GB microSD — local image buffer + OS
Enclosure	3D printed or off-the-shelf project box, mounts to machine interior rear wall

Target BOM cost

Under \$120

COMPUTER VISION APPROACH

How inventory tracking works

We are NOT building a complex ML model for v1. The approach is image differencing — simple, proven, and achievable this summer.

Setup — baseline capture

One time, on installation day

Owner loads machine to full stock. Device takes a high-res baseline photo of the entire product grid. Owner inputs slot layout and prices into the app.

Runtime — periodic comparison

Every 30-60 minutes automatically

Pi wakes, takes a new photo, runs pixel-level comparison against the baseline. Slots where content has changed are flagged. Delta is logged as a sale event and synced to the cloud.

Edge cases — uncertainty flagging

Don't guess, flag it

If a slot is ambiguous (partial depletion, reflection, lighting shift) the system flags it as needs check rather than reporting wrong data. Accuracy over false confidence.

V2 — smarter detection

Post-MVP, if needed

Fine-tune a YOLO model on real machine photos to identify products by name and count partial stock. Only worth pursuing after v1 is in real machines.

WHO OWNS WHAT

TECHNICAL CO-FOUNDER

- Hardware selection and sourcing
- Physical enclosure design and 3D printing
- Raspberry Pi setup and firmware
- Camera mount and angle optimization

BUSINESS CO-FOUNDER — ETHAN

- Product vision and roadmap
- Customer discovery and sales
- App UX decisions and feedback
- Pricing and business model

- Image differencing logic (Python)
- Cloud sync from device to backend
- Power wiring and safety
- Backend API and database
- All technical documentation

- Finding beta machine owners
- Marketing and distribution
- Investor and accelerator outreach
- Brand and naming
- Revenue tracking features spec

SUMMER BUILD TIMELINE

Weeks 1–2	Source hardware, set up Pi, get camera feed running
Weeks 3–4	Build image differencing logic, test on a real machine
Weeks 5–6	Design and print enclosure, finalize power solution
Weeks 7–8	Cloud sync + basic app dashboard, first real install
Weeks 9–10	Bug fixes, accuracy tuning, push notifications

DEFINITION OF DONE

MVP success criteria

One device is installed inside one real vending machine. The owner can open their phone and see a live inventory grid showing which slots are full, low, or empty — without leaving their house. Revenue estimate updates automatically as items sell. That's the finish line for summer.